

Diffygons—Solutions

Fall 2019 ARML Power Contest

Problem 1. Simply apply the difference operation and compute the results:

- (a) $\Delta(9, 17, 31, 57) = (8, 14, 26, 48)$, $\Delta^2(9, 17, 31, 57) = (6, 12, 22, 40)$, $\Delta^3(9, 17, 31, 57) = (6, 10, 18, 34)$, and finally $\Delta^4(9, 17, 31, 57) = \boxed{(4, 8, 16, 28)}$.
 (b) Continuing, $\Delta^5(9, 17, 31, 57) = (4, 8, 12, 24)$, $\Delta^6(9, 17, 31, 57) = (4, 4, 12, 20)$, $\Delta^7(9, 17, 31, 57) = (0, 8, 8, 16)$, $\Delta^8(9, 17, 31, 57) = (8, 0, 8, 16)$, $\Delta^9(9, 17, 31, 57) = (8, 8, 8, 8)$ and finally $\Delta^{10}(9, 17, 31, 57) = (0, 0, 0, 0)$ so the longevity is $\boxed{10}$.

Problem 2.

- (a) $\Delta(4, 2, 8, 3, 6) = \boxed{(2, 6, 5, 3, 2)}$.
 (b) $3 \cdot A + 7 = (12, 6, 24, 9, 18) + 7 = \boxed{(19, 13, 31, 16, 25)}$.
 (c) $\Delta(19, 13, 31, 16, 25) = \boxed{(6, 18, 15, 9, 6)}$.

Problem 3. These follow easily from properties of arithmetic and absolute value.

- (a) Given any two real numbers a and b , note that $(a + c) - (b + c) = a + c + b - c = a - b$ and likewise the absolute values of both sides are equal. Apply this to each slot of ΔA .
 (b) Given any two real numbers a and b , note that $ka - kb = k(a - b)$ and taking absolute values yields $|ka - kb| = |k(a - b)| = |k||a - b|$. Thus, each slot of $\Delta(k \cdot A)$ is simply $|k|$ times the corresponding entry of ΔA , which is the definition of $|k| \cdot \Delta A$.

Problem 4. Let $c = \min(a_1, a_2, \dots, a_n)$ and let $B = A + (-c)$. Since $c \leq a_k$ for any k , $a_k - c \geq 0$ and since $c = a_{\min}$ for whichever subscript(s) correspond to the smallest a , there is at least one zero among the entries of B . So B is an n -tuple whose minimum element is 0, and by problem 3(a) $\Delta B = \Delta A$.

Problem 5.

- (a) Note that adding a constant to all numbers in a list increases or decreases both the minimum and the maximum number in the list by the same amount. Thus, the argument from 3(a) applies to the range.
 (b) Similarly, the argument from 3(b) applies here, since if k is positive both the minimum and maximum are multiplied by the same amount (which then factors out of the difference). If k is negative, the role of the maximum and minimum are interchanged, but then $-k = |k|$ is factored out of the difference. If $k = 0$ then all products become zero as does the range.
 (c) Use part (a) and problem 4 to obtain from A another n -tuple B whose minimum element is zero and whose range is the same as that of A . The maximum element of B must be $r(A)$. But since all the elements of B are nonnegative, the minimum elements of ΔB is

nonnegative and the maximum element of ΔB is at most the maximum element of B . Thus, $r(\Delta A) = r(\Delta B) \leq r(A)$.

Problem 6. To keep track of the relative sizes of the numbers, approximate $\pi \approx 3.1$, $e \approx 2.7$, and $\sqrt{2} \approx 1.4$. So the results of recursively applying Δ to $(\pi, e, 4, \sqrt{2})$ are:

$$\begin{aligned}\Delta(\pi, e, 4, \sqrt{2}) &= (\pi - e, 4 - e, 4 - \sqrt{2}, \pi - \sqrt{2}) \\ \Delta^2(\pi, e, 4, \sqrt{2}) &= (4 - \pi, e - \sqrt{2}, 4 - \pi, e - \sqrt{2}) \\ \Delta^3(\pi, e, 4, \sqrt{2}) &= (a, a, a, a), \text{ where } a = e + \pi - \sqrt{2} - 4 \\ \Delta^4(\pi, e, 4, \sqrt{2}) &= (0, 0, 0, 0)\end{aligned}$$

Problem 7. The table below lists every possible combination of even (e) and odd (o) entries a diffy box could have, and what combination it will have after Δ is applied. Naturally, $o - e$ and $e - o$ both lead to o while $e - e$ and $o - o$ lead to e when computing Δ . Of course, the table is quite redundant (for example, (e, o, o, o) and (o, e, o, o) are just rotations of each other).

A	ΔA	$\Delta^2 A$	$\Delta^3 A$	$\Delta^4 A$
(o, o, o, o)	(e, e, e, e)	(e, e, e, e)	(e, e, e, e)	(e, e, e, e)
(o, o, o, e)	(e, e, o, o)	(e, o, e, o)	(o, o, o, o)	(e, e, e, e)
(o, o, e, o)	(e, o, o, e)	(o, e, o, e)	(o, o, o, o)	(e, e, e, e)
(o, o, e, e)	(e, o, e, o)	(o, o, o, o)	(e, e, e, e)	(e, e, e, e)
(o, e, o, o)	(o, o, e, e)	(e, o, e, o)	(o, o, o, o)	(e, e, e, e)
(o, e, o, e)	(o, o, o, o)	(e, e, e, e)	(e, e, e, e)	(e, e, e, e)
(o, e, e, o)	(o, e, o, e)	(o, o, o, o)	(e, e, e, e)	(e, e, e, e)
(o, e, e, e)	(o, e, e, o)	(o, e, o, e)	(o, o, o, o)	(e, e, e, e)
(e, o, o, o)	(o, e, e, o)	(o, e, o, e)	(o, o, o, o)	(e, e, e, e)
(e, o, o, e)	(o, e, o, e)	(o, o, o, o)	(e, e, e, e)	(e, e, e, e)
(e, o, e, o)	(o, o, o, o)	(e, e, e, e)	(e, e, e, e)	(e, e, e, e)
(e, o, e, e)	(o, o, e, e)	(e, o, e, o)	(o, o, o, o)	(e, e, e, e)
(e, e, o, o)	(e, o, e, o)	(o, o, o, o)	(e, e, e, e)	(e, e, e, e)
(e, e, o, e)	(e, o, o, e)	(o, e, o, e)	(o, o, o, o)	(e, e, e, e)
(e, e, e, o)	(e, e, o, o)	(e, o, e, o)	(o, o, o, o)	(e, e, e, e)
(e, e, e, e)	(e, e, e, e)	(e, e, e, e)	(e, e, e, e)	(e, e, e, e)

Since the entry in the Δ^4 column is always (e, e, e, e) all diffy boxes containing only integers yield all even integers after four applications of Δ .

Problem 8. The idea of the proof is that applying Δ four times, as in the previous problem, leads to all even entries, so 2 can be factored out of the diffy box, *and also out of the range*. But the range stays the same or shrinks when applying Δ . Since the range is finite, factors of two cannot continue to be removed forever!

Specifically, let n be an integer so that $2^n > r(A)$. Define $A_0 = A$. Since $\Delta^4 A_0$ consists of all even terms, define A_1 so that $\Delta^4 A_0 = 2 \cdot A_1$. Recursively define A_k to be such that $\Delta^4 A_{k-1} = 2 \cdot A_k$. Inductively, $\Delta^{4k} A_0 = 2^k \cdot A_k$.

Now consider the range: $2^n > r(A) = r(A_0) \geq r(\Delta^{4n} A_0) = r(2^n A_n) = 2^n r(A_n)$, the last equality courtesy of problem 5(b). Thus, $r(A_n) < 1$ but since all numbers involved are integers the only way this can happen is if $r(A_n) = 0$, meaning all entries of A_n are equal. Then $\Delta A_n = (0, 0, 0, 0)$. Conclusion: $\Delta^{4n+1} A = (0, 0, 0, 0)$.

Problem 9. This will be a brute force computation. Note that the Tribonacci numbers are increasing for indices larger than 1. Then $\Delta(t_n, t_{n+1}, t_{n+2}, t_{n+3}) = (t_{n+1} - t_n, t_{n+2} - t_{n+1}, t_{n+3} - t_{n+2}, t_{n+3} - t_n)$. Applying the recursive definition $t_n = t_{n-1} + t_{n-2} + t_{n-3}$ to these combinations yields $(t_{n-1} + t_{n-2}, t_n + t_{n-1}, t_{n+1} + t_n, t_{n+2} + t_{n+1})$. Note that since indices are as small as $n - 2$ the assumption that $n \geq 2$ was necessary.

Now apply Δ again: $\Delta^2 T_n = (t_n - t_{n-2}, t_{n+1} - t_{n-1}, t_{n+2} - t_n, t_{n+2} + t_{n+1} - t_{n-1} - t_{n-2})$. The last entry may be simplified, by adding and subtracting t_n :

$$\begin{aligned} t_{n+2} + t_{n+1} - t_{n-1} - t_{n-2} &= t_{n+2} + t_{n+1} + t_n - t_n - t_{n-1} - t_{n-2} \\ &= t_{n+2} + t_n \end{aligned}$$

by noting that $t_{n+1} - t_n - t_{n-1} - t_{n-2} = 0$. Thus $\Delta^2 T_n = (t_n - t_{n-2}, t_{n+1} - t_{n-1}, t_{n+2} - t_n, t_{n+2} + t_n)$.

It will simplify the next step to use the recurrence relation to replace $t_{n+1} - t_{n-1}$ with its equivalent $t_n + t_{n-2}$ and $t_{n+2} - t_n$ with $t_{n+1} + t_{n-1}$. Thus $\Delta^2 T_n = (t_n - t_{n-2}, t_n + t_{n-2}, t_{n+1} + t_{n-1}, t_{n+2} + t_n)$.

Apply Δ one more time to produce $\Delta^3 T_n = (2t_{n-2}, t_{n+1} - t_n - t_{n-2} + t_{n-1}, t_{n+2} - t_{n+1} - t_{n-1} + t_n, t_{n+2} + t_n)$. The second and third entries can be simplified by replacing the highest index term using the recurrence relation and simplifying, yielding $2t_{n-1}$ and $2t_n$ respectively. The final entry is simplified as $t_{n+2} + t_{n-2} = t_{n+1} + t_n + t_{n-1} + t_{n-2} = t_{n+1} + t_{n+1} = 2t_{n+1}$, replacing the top index term and then grouping the three lowest-index terms together in the result. The final result is $\Delta^3 T_n = (2t_{n-2}, 2t_{n-1}, 2t_n, 2t_{n+1}) = 2 \cdot T_{n-2}$ as desired.

Problem 10. Compute the longevity of T_0 and T_1 (each arrow represents applying Δ): $T_0 = (0, 0, 1, 1) \rightarrow (0, 1, 0, 1) \rightarrow (1, 1, 1, 1) \rightarrow (0, 0, 0, 0)$ and $T_1 = (0, 1, 1, 2) \rightarrow (1, 0, 1, 2) \rightarrow (1, 1, 1, 1) \rightarrow (0, 0, 0, 0)$. So both these boxes have longevity 3.

Now if n is even, say $n = 2k$, then applying the results of problems 3 and 9 shows that $\Delta^{3k} T_n = 2^k \cdot T_0$ and then three more applications of Δ yields all zeroes. So the longevity is $3k + 3$.

In case n is odd, say $n = 2k + 1$, the result is the $\Delta^{3k} T_n = 2^k T_1$ and, again, three more applications yields all zeroes.

Combining these yields the formula $3n/2 + 3$ for n even and $3(n - 1)/2 + 3$ for n odd. These can be combined in a formula such as $3\lfloor \frac{n}{2} \rfloor + 3$.

Problem 11. Once again, simply compute $\Delta(1, t, t^2, t^3) = (t - 1, t^2 - t, t^3 - t^2, t^3 - 1) = (t - 1, t(t - 1), t^2(t - 1), (t^2 + t + 1)(t - 1)) = (t - 1) \cdot (1, t, t^2, t^2 + t + 1)$. Of course, since $t^3 - t^2 - t - 1 = 0$ the last term in the diffy box can be replaced by t^3 which is the desired result.

Problem 12. Of course, an n -tuple that is all zeroes has a longevity of zero, but no other

n -tuple has longevity zero.

(a) The longevity will be one if and only if applying Δ yields all zeroes, which occurs if and only if all the numbers in the n -tuple are the same.

(b) If the longevity of $A = (a_1, a_2, \dots, a_n)$ is 2, then ΔA has longevity 1, so by part (a) must have the form $(c, c, \dots, c)$ for some non-zero c . Thus $|a_2 - a_1| = c$ or $a_2 - a_1 = \pm c$. Similarly, $a_3 - a_2 = \pm c$, and so forth until $a_n - a_{n-1} = \pm c$ and $a_1 - a_n = \pm c$.

Add all n of these equations. On the left-hand side, each a_i occurs once with a plus and once with a minus, so the left-hand side is zero. On the right-hand side there will be k $+$'s and $n - k$ $-$'s. So the result of adding the $\pm c$'s on the right-hand side is $(2k - n)c$. But c is not zero, and since n is odd $2k - n$ cannot be zero either. Thus the right-hand side cannot equal zero, forcing a contradiction. So no odd-length n -tuple can have longevity equal to 2.

Of course, since the longevity of ΔA is one less than the longevity of A , no odd-length n -tuple can have longevity longer than 2 either.

Problem 13. To make the following simpler, consider any diffy box where two non-adjacent elements are equal, which can be rotated to look like (a, b, a, c) . Start applying Δ : $\Delta(a, b, a, c) = (x, x, y, y)$ where $x = |b - a|$ and $y = |c - a|$. Then $\Delta^2(a, b, a, c) = (0, z, 0, z)$ where $z = |x - y|$. Of course, $\Delta^3(a, b, a, c) = (z, z, z, z)$ and $\Delta^4(a, b, a, c) = (0, 0, 0, 0)$. So if two non-adjacent elements are equal, the longevity is at most four.

From earlier work, only $(0, 0, 0, 0)$ has longevity zero, and only (c, c, c, c) with $c \neq 0$ has longevity 1. All other diffy boxes must have larger longevity. Thus problems 3 and 4 apply without changing the longevity. The diffy box can be rotated so that its minimum element comes first. The results of problems 3 and 4 can then be applied so that this first entry can be assumed to be zero and the first non-zero entry can be assumed to be 1.

The possibilities for diffy boxes are thus scaled, translated, and/or rotated versions of one of the following cases:

$(0, 0, 0, 1)$ This category includes all boxes where the smallest number occurs three times.

Applying Δ yields $(0, 0, 0, 1) \rightarrow (0, 0, 1, 1) \rightarrow (0, 1, 0, 1) \rightarrow (1, 1, 1, 1) \rightarrow (0, 0, 0, 0)$ so if three of the entries are equal the longevity is four.

$(0, 0, 1, 1)$ From the previous case, diffy boxes with this pattern are seen to have longevity 3.

$(0, 0, 1, a)$ If $a > 1$ then this case yields $(0, 0, 1, a) \rightarrow (0, 1, a - 1, a) \rightarrow (1, |(2 - a)|, 1, a)$.

If $a < 1$ then this case yields $(0, 0, 1, a) \rightarrow (0, 1, 1 - a, a) \rightarrow (1, a, |(2a - 1)|, a)$. For both possibilities, two applications of Δ led to a box with equal non-adjacent terms, which has already been shown to take at most four more steps to reach all zeroes.

$(0, 1, 0, a)$ For any a the longevity is at most four because of the two non-adjacent equal entries.

$(0, 1, 1, 0)$ Has longevity 3 from the second case.

$(0, 1, 1, 1)$ Applying Δ yields $(1, 0, 0, 1)$ which is again the second case, so diffy boxes where the largest number occurs three times have longevity four.

$(0, 1, 1, a)$ The evolution is $(0, 1, 1, a) \longrightarrow (1, 0, |a - 1|, a) \longrightarrow (1, |a - 1|, x, |a - 1|)$ where the value of x doesn't matter because two non-adjacent entries are equal, so this case has longevity at most six.

$(0, 1, a, 0)$ This is rotated from $(0, 0, 1, a)$ above.

$(0, 1, a, 1)$ This has two non-adjacent equal entries.

$(0, 1, a, a)$ This evolves as $(0, 1, a, a) \longrightarrow (1, |a - 1|, 0, a) \longrightarrow (x, |a - 1|, a, |a - 1|)$ so this case has longevity at most six, regardless of the value of x .

$(0, 1, a, b)$ Since the terms aren't allowed to increase, either $a < 1$ or $b < a$. *But not both!* Otherwise the terms would strictly increase if we went around the box in the other direction.

In the case $a < 1$, the evolution will be $(0, 1, a, b) \longrightarrow (1, 1 - a, b - a, b) \longrightarrow (a, |b - 1|, a, |b - 1|)$ which lasts at most two more steps. If $a > 1$ and $a > b$ then $(0, 1, a, b) \longrightarrow (1, a - 1, a - b, b) \longrightarrow (|a - 2|, |b - 1|, |a - 2b|, |b - 1|)$ which lasts at most four more steps.

Since all possibilities have now been accounted for, the conclusion that any box where the terms do not strictly increase must have longevity at most six is proven.